CLASS 10 — MATHS · FORMULA SHEET

Chapter 6: **Triangles** | Quick revision, one page

Similar triangles

- Similar: corresponding angles equal and corresponding sides in the same ratio.
- **AA** (two angles equal), **SSS** (three pairs of sides proportional), **SAS** (one equal angle between two proportional sides).
- All circles are similar, all squares are similar, all equilateral triangles are similar.

Basic Proportionality Theorem (Thales)

- If $DE \parallel BC$ in $\triangle ABC$ (D on AB, E on AC): $\dfrac{AD}{DB} = \dfrac{AE}{EC}$.
- Converse: if $\dfrac{AD}{DB} = \dfrac{AE}{EC}$ then $DE \parallel BC$.
- Also $\dfrac{AD}{AB} = \dfrac{AE}{AC} = \dfrac{DE}{BC}$ when $DE \parallel BC$.

Areas of similar triangles

- $\dfrac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle PQR)} = \left(\dfrac{AB}{PQ}\right)^2 = \left(\dfrac{BC}{QR}\right)^2 = \left(\dfrac{CA}{RP}\right)^2$.
- Ratio of areas = square of the ratio of corresponding sides.

Pythagoras theorem and converse

- In a right triangle: $(\text{hypotenuse})^2 = (\text{base})^2 + (\text{perpendicular})^2$.
- Converse: if $c^2 = a^2 + b^2$ then the angle opposite c is 90°.
- Common triples: $(3, 4, 5),\ (5, 12, 13),\ (8, 15, 17),\ (7, 24, 25)$.

Applications

- Shadow problems: $\dfrac{\text{height}}{\text{shadow}}$ is the same for all objects at the same time.
- Altitude to hypotenuse creates triangles similar to the whole triangle.

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